

1. Assalamu Alaikum Program

```
// Header file for input output functions
#include <stdio.h>
// Main function: entry point for execution
int main(){
    // Writing print statement to print hello world
    printf("Assalamu Alaikum ");
    return 0;
}
```

2. C Program to Print Your Own Name

```
#include <stdio.h>
int main() {
    // Defining string (character array) assuming 100
    char name[100];
    printf("Enter Your Name: ");
    scanf("%s", name);
    printf("Your Name: %s\n", name);
    return 0;
}
```

3. C Program to Print an Integer Entered By the User

```
#include <stdio.h>
int main(){
    // Declare the variables
    int num;
    printf("Enter the integer: ");
    scanf("%d", &num);
    printf("Entered integer is: %d", num);
    return 0;
}
```

4. C Program to Add Two Numbers

```
#include <stdio.h>
int main() {
    int a, b, sum = 0;
    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);
    sum = a + b;
```

```

printf("Sum: %d", sum);
return 0;
}

```

// C program to add two numbers using Bitwise operators

```

#include <stdio.h>
int main() {
    int a, b, sum, carry;
    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);
    while (b) {
        // Carry is AND of a and b
        carry = a & b;
        // Sum without carry is XOR of a and b
        a = a ^ b;
        // Carry is shifted by one so that it can be
        // added in the next iteration
        b = carry << 1;
    }
    printf("Sum = %d\n", a);
    return 0;
}

```

5. C Program to Check Whether a Number is Prime or Not

Square and square root method apply
 //Brute Force Method - O(n) Time

```

#include <stdbool.h>
#include <stdio.h>
int main() {
    int n = 29;
    int cnt = 0;
    // If number is less than/equal to 1,
    // it is not prime
    if (n <= 1)
        printf("%d is NOT prime", n);
    else {
        // Count the all divisors of
        // given number
        for (int i = 1; i <= n; i++) {
            // Check n is divided by
            // i or not
            if (n % i == 0)
                cnt++;
        }
    }
}

```

```

// If n is divisible by more than 2 numbers
// then it is not prime
if (cnt > 2)
    printf("%d is NOT prime", n);
// else it is prime
else
    printf("%d is prime", n);
}
return 0;
}

```

6. C Program to Multiply two Floating-Point Numbers

```

#include <stdio.h>
float multiply(float a, float b)
{
    return a * b;
}
int main()
{
    float A = 2.12, B = 3.88, product;
    product = multiply(A, B);
    printf("Product of entered numbers is: %.3f", product);
    return 0;
}

```

7. C Program to Print the ASCII Value of a Character

```

// typecasting into integer //Explicit
#include <stdio.h>
int main() {
    char ch = 'A';
    // Find the ASCII value of a character using typecasting
    int asciiValue = (int)ch;
    printf("ASCII value of %c is %d\n", ch, asciiValue);
    return 0;
}
// implicit conversion with format specifier.
#include <stdio.h>
int main() {
    char c = 'k';
    printf("The ASCII value of %c is %d", c, c);
    return 0;
}

```

8. C Program to Swap Two Numbers

```
#include <stdio.h>
int main() {
    int a = 5, b = 10, temp;
    temp = a;
    a = b;
    b = temp;
    printf("a = %d, b = %d\n", a, b);
    return 0;
}
```

//Without temporary variable

```
#include <stdio.h>
int main() {
    int a = 5, b = 10;
    a = a + b;
    b = a - b;
    a = a - b;
    printf("a = %d, b = %d\n", a, b);
    return 0;
}
```

~~9. C~~ Program to Calculate Fahrenheit to Celsius

```
#include <stdio.h>
float fahrenheit_to_celsius(float f)
{
    return ((f - 32.0) * 5.0 / 9.0);
}
int main()
{
    float f = 40;
    printf("Temperature in Degree Celsius : %.2f",
        fahrenheit_to_celsius(f));
    return 0;
}
```

~~10. C~~ Program to Find the Size of int, float, double, and char

```
#include <stdio.h>
int main() {
    printf("Size of int: %u bytes\n", sizeof(int));
    printf("Size of float: %u bytes\n", sizeof(float));
    printf("Size of double: %u bytes\n", sizeof(double));
    printf("Size of char: %u bytes\n", sizeof(char));
    return 0;
}
```

~~11.~~ C Program to Add Two Complex Numbers

```
#include <stdbool.h>
```

```
#include <stdio.h>
```

```
#include <math.h>
```

```
// This function is to check
```

```
// if a given number is prime
```

```
bool isPrime(int n)
```

```
{
```

```
    // 0 and 1 are not prime numbers
```

```
    if (n == 1 || n == 0)
```

```
        return false;
```

```
    // Check for divisibility from 2 to sqrt(n)
```

```
    for (int i = 2; i <= sqrt(n); i++) {
```

```
        if (n % i == 0)
```

```
            return false;
```

```
    }
```

```
    return true;
```

```
}
```

```
// Driver code
```

```
int main()
```

```
{
```

```
    int N = 50;
```

```
    // Check every number from 1 to N
```

```
    for (int i = 1; i <= N; i++) {
```

```
        if (isPrime(i)) {
```

```
            printf("%d ", i);
```

```
        }
```

```
    }
```

```
    return 0;
```

```
}
```

PR

12. C Program to Print Prime Numbers From 1 to N

```
#include <stdbool.h>
```

```
#include <stdio.h>
```

```
#include <math.h>
```

```
// This function is to check
```

```
// if a given number is prime
```

```

bool isPrime(int n)
{
    // 0 and 1 are not prime numbers
    if (n == 1 || n == 0)
        return false;

    // Check for divisibility from 2 to sqrt(n)
    for (int i = 2; i <= sqrt(n); i++) {
        if (n % i == 0)
            return false;
    }
    return true;
}

```

```

// Driver code
int main()
{
    int N = 50;

    // Check every number from 1 to N
    for (int i = 1; i <= N; i++) {
        if (isPrime(i)) {
            printf("%d ", i);
        }
    }
    return 0;
}

```

13. C Program to Find Simple Interest

```

#include <stdio.h>

int main() {
    float P = 1, R = 1, T = 1;
    float SI = (P * T * R) / 100;
    printf("Simple Interest = %f\n", SI);
    return 0;
}

```

// C Program to Calculate Simple Interest using Functions

```

#include <stdio.h>

float smpInt(float p, float r, float t) {
    return (p * r * t) / 100;
}

int main() {
    float P = 10000, R = 12, T = 1, SI;

```

```

    SI = smpInt(P, R, T);
    printf("Simple Interest: %.2f\n", SI);
    return 0;
}

```

14. C Program to Find Compound Interest

```

#include <stdio.h>
#include<math.h>
int main(){
    double principal = 10000;
    double rate = 5;
    double time = 2;
    double Amount = principal * ((pow((1 + rate / 100), time)));
    double CI = Amount - principal;
    printf("Compound Interest is : %lf",CI);
    return 0;
}

```

15. C Program for Area And Perimeter Of Rectangle

```

#include <stdio.h>
int main(){
    int l = 10, b = 10;
    int A, P;
    A = l * b;
    P = 2 * (l + b);
    printf("Area of rectangle is : %d", A);
    printf("\nPerimeter of rectangle is : %d", P);
    return 0;
}

```

1. C Program to Check Whether a Number is Positive, Negative, or Zero

```

#include <stdio.h>
void checkNum(int N) {
    if (N == 0) {
        printf("Zero\n");
    }
    else if (N < 0) {
        printf("Negative\n");
    }
    else {
        printf("Positive\n");
    }
}
int main() {

```

```
int N = 10;
checkNum(N);
return 0;
}
```

2. C Program to Check Whether Number is Even or Odd

// C Program to Check Even or Odd Using Bitwise

// AND Operator

#include <stdio.h>

void checkEvenOdd(int N) {

if (N & 1) {
 printf("Odd\n");

}

else {
 printf("Even\n");
 }

}

int main() {

int N = 7;

checkEvenOdd(N);

return 0;

}

3. C Program to Check Whether a Character is Vowel or Consonant

#include <stdio.h>

int main()s{

char ch = 'A';

if (ch == 'a' || ch == 'A' || ch == 'e' || ch == 'E'
 || ch == 'i' || ch == 'I' || ch == 'o' || ch == 'O'
 || ch == 'u' || ch == 'U')

{

printf("The character %c is a vowel.\n", ch);

}

else {

printf("The character %c is a consonant.\n", ch);

}

return 0;

}

4. C Program to Find Largest Number Among Three Numbers

#include <stdio.h>

int main(){

int c = 10, b = 22, a = 9;

if (a >= b) {

if (a >= c)

printf("%d is the largest number.", a);

else

```

        printf("%d is the largest number.", c);
    }
    else {
        if (b >= c)
            printf("%d is the largest number.", b);
        else
            printf("%d is the largest number.", c);
    }
    return 0;
}

```

// Using if-else statement

```
#include <stdio.h>
```

```

int main() {
    int a = 11, b = 2, c = 9;

    // Finding max using compound expressions
    if (a > b && a > c)
        printf("%d is the largest number.", a);
    else if (b > a && b > c)
        printf("%d is the largest number.", b);
    else if (c > a && c > b)
        printf("%d is the largest number.", c);
    else
        printf("THREE NUMBER ARE EQUAL");
    return 0;
}

```

// Temporary Variable

```
#include <stdio.h>
```

```

int main() {
    int a = 10, b = 22, c = 9;
    int max = a;
    if (max < b)
        max = b;
    if (max < c)
        max = c;
    printf("%d is the largest number.", max);
    return 0;
}

```

// Using a Custom Max Function

```
#include <stdio.h>
```

```

int findMax(int b, int a) {
    return (a >= b) ? a : b;
}

```

```

int main() {
    int a = 10, b = 22, c = 9;

    int max = findMax(max(a, b), c);
    printf("%d is the largest number.", max);
    return 0;
}

```

5. C Program to Calculate Sum of Natural Numbers

```
#include <stdio.h>
```

```

int main(){
    int i, s = 0;
    int n = 10;
    i = 1;
    while (i <= n) {
        s += i;
        i++;
    }
    printf("Sum = %d", s);
    return 0;
}

```

// using for loops

```
#include <stdio.h>
```

```

int main(){

    int i, s = 0;
    int n = 10;

    for (i = 0; i <= n; i++) {
        s += i;
    }
    printf("Sum = %d", s);
    return 0;
}

```

// using functions

```
#include <stdio.h>
```

```

int sumofnaturalnumbers(int num){
    int i, s = 0;
    for (i = 0; i <= num; i++) {
        s += i;
    }
    printf("Sum = %d", s);
}
int main()
{

```

```

int number = 10;
sumofnaturalnumbers(number);
return 0;
}

```

```

// Using formula
#include <stdio.h>
int main(){
    int num = 10;
    int s,x;
    s=num*(num+1);
    x=(int)(s/2);
    printf("Sum = %d", x);
    return 0;
}

```

6. C Program to Print Alphabets From A to Z Using Loop

```

#include <stdio.h>
int main(){
    int i;
    printf("Alphabets from (A-Z) are:\n");
    for (i = 65; i <= 90; i++) {
        printf("%c ", i);
    }
    printf("\nAlphabets from (a-z) are:\n");
    for (i = 97; i <= 122; i++) {
        printf("%c ", i);
    }
    return 0;
}

```

```

// Using Character and for loop
#include <stdio.h>
int main(){
    char i;
    printf("The Alphabets from A to Z are: \n");
    for (i = 'A'; i <= 'Z'; i++) {
        printf("%c ", i);
    }
    printf("\nThe Alphabets from a to z are: \n");

    for (i = 'a'; i <= 'z'; i++) {
        printf("%c ", i);
    }
    return 0;
}

```

```
// (a to z) using while loop
#include <stdio.h>
int main(){
    char i;
    printf("The Alphabets from A to Z are: \n");
    i = 'A';
    while (i <= 'Z') {
        printf("%c ", i);
        i++;
    }

    // for lower case alphabets
    printf("\nThe Alphabets from a to z are: \n");
    i = 'a';
    while (i <= 'z') {
        printf("%c ", i);
        i++;
    }
    return 0;
}
```

```
// (a to z) using do-while loop
#include <stdio.h>
int main()
{
    char i;
    printf("The Alphabets from A to Z are: \n");

    i = 'A';
    do
    {
        printf("%c ", i);
        i++;
    } while (i <= 'Z');

    // for lower case alphabets
    printf("\nThe Alphabets from a to z are: \n");

    i = 'a';

    do {
        printf("%c ", i);
        i++;
    } while (i <= 'z');

    return 0;
}
```

7. C Program to Check Leap Year

```
#include<stdio.h>
int main()
{
    int y;
    printf("Enter your year : ");
    scanf("%d",&y);
    if (y%400==0)
        printf("\nYour %d is LEAP year\n",y);
    else if (y%4==0 && y%100!=0)
        printf("\nYour %d is LEAP year\n",y);
    else
        printf("\nYour %d is NOT LEAP year\n",y);

    return 0;
}
```

8. C Program to Find Factorial of a Number

```
#include <stdio.h>
int factorial(int N) {
    int fact = 1, i;
    for (i = 1; i <= N; i++)
    {
        fact *= i;
    }
    return fact;
}

int main() {
    int N = 5;
    int fact = factorial(N);
    printf("Factorial of %d is %d", N, fact);
    return 0;
}
```

9. C Program to Make a Simple Calculator

```
// switch-case statements
#include <stdio.h>
int main() {
    char op;
    double a, b, res;
    printf("Enter an operator (+, -, *, /): ");
    scanf("%c", &op);
    printf("Enter two operands: ");
    scanf("%lf %lf", &a, &b);
```

```

switch (op) {
case '+':
    res = a + b;
    break;
case '-':
    res = a - b;
    break;
case '*':
    res = a * b;
    break;
case '/':
    res = a / b;
    break;
default:
    printf("Error! Incorrect Operator Value\n");
}
return 0;
}

```

// using if-else statements

```
#include <stdio.h>
```

```

int main() {
    char op;
    double a, b, res;
    printf("Enter an operator (+, -, *, /): ");
    scanf("%c", &op);
    printf("Enter two operands: ");
    scanf("%lf %lf", &a, &b);

    if (op == '+')
        res = a + b;
    else if (op == '-')
        res = a - b;
    else if (op == '*')
        res = a * b;
    else if (op == '/')
        res = a / b;
    else {
        printf("Error! Operator is not correct.\n");
    }
    return 0;
}

```

10. C Program to Generate Multiplication Table

```
#include <stdio.h>
```

```

void print_table(int range, int num)
{
    int mul;
    for (int i = 1; i <= range; i++)
    {
        mul = num * i;
        printf("%d * %d = %d", num, i, mul);
        printf("\n");
    }
}

int main()
{
    int range = 10;
    int num = 5;
    print_table(range, num);
    return 0;
}

```

11. C Program to Print Fibonacci Series

The Fibonacci series is the sequence where each number is the sum of the previous two numbers of the sequence. The first two numbers are 0 and 1 which are used to generate the whole series. output is 0 1 1 2 3 5 8 13 21

```

#include <stdio.h>
void printFib(int n)
{
    if (n < 1)
    {
        printf("Invalid Number of terms\n");
        return;
    }
    int prev1 = 1;
    int prev2 = 0;
    for (int i = 1; i <= n; i++) {
        if (i > 2)
        {
            int curr = prev1 + prev2;
            prev2 = prev1;
            prev1 = curr;
            printf("%d ", curr);
        }
        else if (i == 1)
            printf("%d ", prev2);
        else if (i == 2)
            printf("%d ", prev1);
    }
}

```

```

int main() {
    int n = 9;
    printFib(n);
    return 0;
}

// Recursive function to print the fibonacci series
#include <stdio.h>
void fib(int n, int prev1, int prev2) {
    if (n < 3) {
        return;
    }
    int curr = prev1 + prev2;
    printf("%d ", curr);

    return fib(n - 1, prev2, curr);
}

// Function that handles the first two terms and calls the recursive function
void printFib(int n) {
    if (n < 1) {
        printf("Invalid number of terms\n");
    }
    else if (n == 1) {
        printf("%d ", 0);
    }
    else if (n == 2) {
        printf("%d %d", 0, 1);
    }
    else {
        printf("%d %d ", 0, 1);
        fib(n, 0, 1);
    }
    return;
}

int main() {
    int n = 9;

    // Printing first 9 Fibonacci series terms
    printFib(n);

    return 0;
}

```

12. C Program to Find LCM of Two Numbers

```
#include <stdio.h>
```

/ GCD

$$a \times b = \text{LCM} \times \text{GCD}$$

```
int main()
{
    int x = 15, y = 25, max;
    max = (x > y) ? x : y;
    while (1) {
        if (max % x == 0 && max % y == 0) {
            printf("The LCM of %d and %d is %d.", x, y, max);
            break;
        }
        ++max;
    }
    return 0;
}

//
```

13. C Program to Check Armstrong Number

An Armstrong number is defined as a number that is equal to the sum of the K^{th} power of each digit in the number, where K is the number of digits in it.

Input: 153

Output: Yes

Explanation: 153 is an Armstrong number of 3 digits, since the sum of cubes of each digit is equal to the number itself.

$$1^3 + 5^3 + 3^3 = 153$$

Input: 12

Output: No

Explanation: 12 is not an Armstrong number as the sum of square of each digit is not equal to the number:

$$1^2 + 2^2 = 5 \neq 12$$

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {
```

```
    int num, originalNum, remainder, result = 0, n = 0;
```

```
    printf("Enter an integer: ");
```

```
    scanf("%d", &num);
```

```
    originalNum = num;
```

```
    // Count the number of digits
```

```
    while (originalNum != 0) {
```

```
        originalNum /= 10;
```

```
        n++;
```

```

}
originalNum = num;

// Calculate the sum of digits raised to the power n
while (originalNum != 0) {
    remainder = originalNum % 10;
    result += pow(remainder, n);
    originalNum /= 10;
}
if (result == num)
    printf("%d is an Armstrong number.\n", num);
else
    printf("%d is not an Armstrong number.\n", num);
return 0;
}

```

14. C Program to Display Armstrong Numbers Between 1 to 1000

15. C Program to Display Armstrong Number Between Two Intervals

```
#include <stdio.h>
```

```
#include <math.h>
```

```

int main() {
    int lower, upper, num, originalNum, remainder, result, n;
    printf("Enter lower limit: ");
    scanf("%d", &lower);
    printf("Enter upper limit: ");
    scanf("%d", &upper);
    printf("Armstrong numbers between %d and %d are:\n", lower, upper);

    for(num = lower; num <= upper; num++) {
        originalNum = num;
        result = 0;

        // Count number of digits
        n = 0;
        while (originalNum != 0) {
            originalNum /= 10;
            n++;
        }

        originalNum = num;

        // Sum of digits raised to power n
        while (originalNum != 0) {
            remainder = originalNum % 10;
            result += pow(remainder, n);
            originalNum /= 10;
        }
    }
}

```

```

    }

    if (result == num)
        printf("%d\n", num);
    }
    return 0;
}

```

16. C Program to Reverse a Number

```

#include <stdio.h>
int reverseDigits(int num)
{
    int rev_num = 0;
    while (num > 0) {
        rev_num = rev_num * 10 + num % 10;
        num = num / 10;
    }
    return rev_num;
}

// Driver code
int main()
{
    int num = 4562;
    printf("Reverse of is %d", reverseDigits(num));
    return 0;
}

```

17. C Program to Check Whether a Number is a Palindrome or Not

Input: 121

Output: Yes

Explanation: The number 121 remains the same when its digits are reversed.

Input: 123

Output: No

Explanation: The number 123 does not remain the same when its digits are reversed.

```

#include <stdio.h>
int main() {
    int num, originalNum, reversed = 0, remainder;
    printf("Enter a number: ");
    scanf("%d", &num);

    originalNum = num;
    // reverse num
    while (num != 0) {

```

```

    remainder = num % 10;
    reversed = reversed * 10 + remainder;
    num /= 10;
}
if (originalNum == reversed)
    printf("%d is a Palindrome number.\n", originalNum);
else
    printf("%d is not a Palindrome number.\n", originalNum);
return 0;
}

```

18. C Program to Check Whether a Number is Prime or Not

```

#include <stdio.h>
int main() {
    int num, i, isPrime = 1;

    printf("Enter a positive number: ");
    scanf("%d", &num);

    // 0 or 1 no Prime
    if (num <= 1) {
        printf("%d is not a Prime number.\n", num);
        return 0;
    }

    // 2 to num before check
    for(i = 2; i <= num / 2; i++) {
        if(num % i == 0) {
            isPrime = 0;
            break;
        }
    }

    if (isPrime)
        printf("%d is a Prime number.\n", num);
    else
        printf("%d is not a Prime number.\n", num);

    return 0;
}

```

19. C Program to Display Prime Numbers Between Intervals

```

#include <stdio.h>
int main() {
    int start, end, i, j, isPrime;
    printf("Enter starting number: ");

```

```

scanf("%d", &start);
printf("Enter ending number: ");
scanf("%d", &end);

printf("Prime numbers between %d and %d are:\n", start, end);

for(i = start; i <= end; i++)
{
    if(i < 2)
        continue; // Skip numbers less than 2

    isPrime = 1; // Assume current number is prime

    // Check if i is divisible by any number between 2 and i-1
    for(j = 2; j <= i/2; j++) {
        if(i % j == 0) {
            isPrime = 0;
            break;
        }
    }
    // If isPrime is still 1, then it's a prime number
    if(isPrime == 1) {
        printf("%d ", i);
    }
}
return 0;
}

```

20. C Program to Check whether the input number is a Neon Number

Given a number (num) we need to check whether it is a Neon Number (i.e. a number where the sum of digits of the square of the number is equal to the number) and return "true" or "false" accordingly.

Example:

Input: num = 9

Output: true

Explanation: square of 9 is $9 * 9 = 81$, sum of digit of square is $8 + 1 = 9$ (i.e equal to given number).

Input: num = 10

Output: false

Explanation: Square of 10 is $10 * 10 = 100$, sum of digit of square is $1 + 0 + 0 = 1$ (i.e. not equal to given number).

```
#include <stdio.h>
```

```
int main() {
```

```
    int num, square, sum = 0;
```

```
    // Input number
```

```

printf("Enter a number: ");
scanf("%d", &num);

// Find square of the number
square = num * num;

// Calculate sum of digits of the square
while(square > 0) {
    sum += square % 10;
    square /= 10;
}

// Check if sum equals the original number
if(sum == num)
    printf("%d is a Neon Number.\n", num);
else
    printf("%d is not a Neon Number.\n", num);
return 0;
}

```

21. C Program to Find All Factors of a Natural Number

```

#include <stdio.h>
int main() {
    int num, i;
    printf("Enter a number: ");
    scanf("%d", &num);

    printf("Factors of %d are:\n", num);
    for(i = 1; i <= num; i++) {
        if(num % i == 0) {
            printf("%d ", i);
        }
    }
    return 0;
}

```

output:

```

Enter a number: 12
Factors of 12 are:
1 2 3 4 6 12

```

22. C program to Sum of Fibonacci Numbers at Even Indexes up to N Terms

```

#include <stdio.h>
int main() {
    int n, i;
    int a = 0, b = 1, next;

```

```

int sum = 0;

printf("Enter the number of Fibonacci terms: ");
scanf("%d", &n);

printf("Fibonacci series: ");

for(i = 0; i < n; i++) {
    if(i == 0) {
        next = 0;
    } else if(i == 1) {
        next = 1;
    } else {
        next = a + b;
        a = b;
        b = next;
    }
    printf("%d ", next);
    if(i % 2 == 0) { // check even index
        sum += next;
    }
}

printf("\nSum of Fibonacci numbers at even indexes = %d\n", sum);
return 0;
}

```

output:

Enter the number of Fibonacci terms: 7

Fibonacci series: 0 1 1 2 3 5 8

Sum of Fibonacci numbers at even indexes = 11

→ Because the even index terms are: 0th(0), 2nd(1), 4th(3), 6th(8)

Sum = 0 + 1 + 3 + 8 = 12

C Program - Pattern Printing

C Program to Print Simple Pyramid Pattern

C Program to Print Given Triangle

C Program to Print 1800 Rotation of Simple Pyramid

C Program to Print Inverted Pyramid

C Program to Print Number Pattern

C Program to Print Character Pattern
C Program to Print Continuous Character Pattern
C Program to Print Hollow Star Pyramid
C Program to Print Inverted Hollow Star pyramid
C Program to Print Hollow Star Pyramid in a Diamond Shape
C Program to Print Full Diamond Shape Pyramid
C Program to Print Pascal's Pattern Triangle Pyramid
C Program to Print Floyd's Pattern Triangle Pyramid
C Program to Print Reverse Floyd pattern Triangle Pyramid